

	CMJD - Comprehensive Master Java Developer Course work – Spring Boot
	CMJD106/107

Assignment 1: Backend Development with Spring Boot and MySQL

Project: Point of Sale System (POS)

Objective: Develop the backend of a Point of Sales System using Spring Boot and MySQL. The system should include functionalities for managing items, item categories, stock, and point of sale. Additionally, implement authentication using Spring Security with JWT.

Tasks:

1. Spring Boot Setup:

- Create a new Spring Boot project.
- Configure MySQL database connectivity.

2. Entity and Repository:

- Define entities for Item, ItemCategory, and Stock.
- Implement JPA repositories for CRUD operations.

3. Authentication:

- Implement JWT authentication using Spring Security.
- Create endpoints for user registration and login.

4. Item and Item Category Management:

- Create REST API endpoints for managing items (add, edit, delete).
- Implement endpoints for managing item categories.

5. Stock Management:

- Develop endpoints for managing stock (add, update, retrieve).
- Ensure proper validation to handle stock transactions.

6. Point of Sale:

- Design REST APIs for handling point of sale transactions.
- Include functionalities like adding items to the cart, generating invoices, and updating stock.

7. **Documentation:**

- Document the API endpoints using Swagger or any suitable documentation tool.

8. **Testing:**

- Write unit tests for critical components, especially authentication and transaction-related functionalities.